Bayesian Optimization Report: Function 3

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.733	Good
2	0.732	Good
3	0.732	Good
4	0.731	Good
5	0.731	Good
6	0.729	Good
7	0.728	Good
8	0.728	Good
9	0.727	Good
10	0.727	Good
11	0.724	Good
12	0.725	Good
13	0.725	Good
14	0.721	Good
15	0.722	Good
16	0.722	Good
17	0.722	Good
18	0.723	Good
19	0.723	Good
20	0.723	Good
21	0.720	Good
22	0.720	Good
23	0.720	Good
24	0.721	Good
25	0.721	Good
26	0.722	Good
27	0.723	Good
28	0.720	Good
29	0.721	Good
30	0.719	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.1928	0.0610	0.3745
2	0.1632	0.0472	0.3152
3	0.1517	0.0663	0.2134
4	0.1474	0.0583	0.2260
5	0.1342	0.0491	0.2056
6	0.1222	0.0541	0.1873
7	0.1178	0.0462	0.1706
8	0.1141	0.0554	0.1633
9	0.1109	0.0652	0.1556
10	0.1069	0.0581	0.1505
11	0.1034	0.0479	0.1999
12	0.1001	0.0543	0.1818
13	0.0939	0.0357	0.1575
14	0.0910	0.0349	0.1899
15	0.0882	0.0458	0.1479
16	0.0872	0.0354	0.1477
17	0.0860	0.0403	0.1552
18	0.0860	0.0443	0.1427
19	0.0808	0.0326	0.1396
20	0.0791	0.0334	0.1410
21	0.0760	0.0263	0.1426
22	0.0763	0.0269	0.1376
23	0.0748	0.0361	0.1572
24	0.0713	0.0357	0.1323
25	0.0729	0.0334	0.1412
26	0.0707	0.0253	0.1449
27	0.0732	0.0308	0.1485
28	0.0687	0.0322	0.1380
29	0.0682	0.0281	0.1406
30	0.0700	0.0278	0.1480

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.33457279 0.81317126 0.48880388]	-0.03937498855881083	Matern(ν=0.5)	UCB	0.733	None
2	[0.40757497 0.6294244 0.54203802]	-0.0499871487338337	Matern(ν=0.5)	UCB	0.732	None
3	[0.41813695 0.87982504 0.42313891]	-0.024076638428444763	Matern(ν=0.5)	UCB	0.732	None
4	[0.39661886 0.63405571 0.43993781]	-0.022872772880168313	Matern(ν=0.5)	UCB	0.731	None
5	[0.31024836 0.79388987 0.38635969]	-0.021756047319420713	Matern(ν=0.5)	UCB	0.731	None
6	[0.52666074 0.79783885 0.38458819]	-0.01999930900028457	Matern(ν=0.5)	UCB	0.729	None
7	[0.25140005 0.61885347 0.38227664]	-0.0212898643528181	Matern(ν=0.5)	UCB	0.728	None
8	[0.43241457 0.58209796 0.39523667]	-0.019103616547981206	Matern(ν=0.5)	UCB	0.728	None
9	[0.5833868 0.86050746 0.43076552]	-0.022972838286491104	Matern(ν=0.5)	UCB	0.727	None
10	[0.49407217 0.70916187 0.3979276]	-0.016768158845365422	Matern(ν=0.5)	UCB	0.727	None
11	[0.55389644 0.59945259 0.4101311]	-0.01874580488016897	Matern(ν=0.5)	UCB	0.724	None
12	[0.59799481 0.73271977 0.42529845]	-0.018932427885717917	Matern(ν=0.5)	UCB	0.725	None
13	[0.54606917 0.67748932 0.40360525]	-0.016575620379429612	Matern(ν=0.5)	UCB	0.725	None
14	[0.64358414 0.6095984 0.41254712]	-0.019319654412492948	Matern(ν=0.5)	UCB	0.721	None
15	[0.68713929 0.694203 0.41629537]	-0.018402214240217818	Matern(ν=0.5)	UCB	0.722	None
16	[0.46179765 0.63853696 0.39963824]	-0.017335646097126345	Matern(ν=0.5)	UCB	0.722	None
17	[0.66874697 0.81038914 0.39958819]	-0.0186900053190433	Matern(ν=0.5)	UCB	0.722	None
18	[0.40155154 0.67904455 0.39157286]	-0.017655443088107736	Matern(ν=0.5)	UCB	0.723	None
19	[0.69343868 0.74483124 0.40292836]	-0.01782514530612911	Matern(ν=0.5)	UCB	0.723	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.6003056236226727)}}

20	[0.54278298 0.72480802 0.39745878]	-0.01682061472 3571214	Matern(ν=0.5)	UCB	0.723	None
21	[0.59390246 0.65699587 0.40015188]	-0.01687168164 742414	Matern(ν=0.5)	UCB	0.720	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.604234802570 6925)}}
22	[0.45236188 0.68981777 0.40557221]	-0.01694010408 3713825	Matern(ν=0.5)	UCB	0.720	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.614754658076 7015)}}
23	[0.63310457 0.74334088 0.40198732]	-0.01710349594 7231595	Matern(ν=0.5)	UCB	0.720	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.624268182021 6613)}}
24	[0.58381007 0.70235091 0.40096496]	-0.01646823561 7626004	Matern(ν=0.5)	UCB	0.721	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.618001345444 7658)}}
25	[0.59547481 0.76350917 0.40027408]	-0.01738554233 726397	Matern(ν=0.5)	UCB	0.721	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.612545343333 0313)}}
26	[0.64046177 0.68884557 0.40675184]	-0.01694790517 437661	Matern(ν=0.5)	UCB	0.722	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.611015054695 0286)}}
27	[0.5215565 0.66906562 0.39928008]	-0.01674892829 4544996	Matern(ν=0.5)	UCB	0.723	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.610181568771 2768)}}
28	[0.71541894 0.66038875 0.40371458]	-0.01860020809 5110204	Matern(ν=0.5)	UCB	0.720	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.flo at64(-0.610283622049 0331)}}

29	[0.60161451 0.71219269 0.40249062]	-0.016548202476101892	Matern(ν=0.5)	UCB	0.721	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.6144753696865547)}}
30	[0.61297693 0.68484025 0.40135621]	-0.016524770856111137	Matern(ν=0.5)	UCB	0.719	{'type': 'heteroscedasticity', 'severity': 0.6, 'details': {'corr_residuals': np.float64(-0.6147400690222274)}}

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.738703
EI	0.048087
PI	0.375537
THOMPSON	1.575276
KG	0.048087
ENTROPY	3.116894
PORTFOLIO	1.693778

Iteration 2

Method	Max Value
UCB	1.434954
EI	0.046479
PI	0.388246
THOMPSON	1.560833
KG	0.046479
ENTROPY	2.899821
PORTFOLIO	1.395388

Iteration 3

Method	Max Value
UCB	1.431722
EI	0.048196

PI	0.386588
THOMPSON	1.697548
KG	0.048196
ENTROPY	2.792873
PORTFOLIO	1.379509

Iteration 4

Method	Max Value
UCB	1.420108
EI	0.044866
PI	0.396404
THOMPSON	1.576838
KG	0.044866
ENTROPY	2.756884
PORTFOLIO	1.272941

Iteration 5

Method	Max Value
UCB	1.408333
EI	0.041861
PI	0.404233
THOMPSON	1.470559
KG	0.041861
ENTROPY	2.482056
PORTFOLIO	1.171087

Iteration 6

Method	Max Value
UCB	1.402151
EI	0.040745
PI	0.404518
THOMPSON	1.409955
KG	0.040745
ENTROPY	2.523211
PORTFOLIO	1.159016

Iteration 7

Method	Max Value
UCB	1.399557
EI	0.040474
PI	0.404573
THOMPSON	1.486212
KG	0.040474
ENTROPY	2.459628
PORTFOLIO	1.136447

Iteration 8

Method	Max Value
UCB	1.389757
EI	0.036659
PI	0.392068
THOMPSON	1.444245
KG	0.036659
ENTROPY	2.459640
PORTFOLIO	1.103622

Iteration 9

Method	Max Value
UCB	1.386602
EI	0.035520
PI	0.384016
THOMPSON	1.509812
KG	0.035520
ENTROPY	2.391862
PORTFOLIO	1.073311

Iteration 10

Method	Max Value
UCB	1.382775
EI	0.034697

PI	0.393218
THOMPSON	1.529672
KG	0.034697
ENTROPY	2.444770
PORTFOLIO	1.056903

Iteration 11

Method	Max Value
UCB	1.371663
EI	0.025457
PI	0.364684
THOMPSON	1.422311
KG	0.025457
ENTROPY	2.367531
PORTFOLIO	1.046765

Iteration 12

Method	Max Value
UCB	1.368886
EI	0.025033
PI	0.362796
THOMPSON	1.412718
KG	0.025033
ENTROPY	2.417313
PORTFOLIO	1.028530

Iteration 13

Method	Max Value
UCB	1.365566
EI	0.023447
PI	0.354766
THOMPSON	1.439114
KG	0.023447
ENTROPY	2.274091
PORTFOLIO	1.008438

Iteration 14

Method	Max Value
UCB	1.347513
EI	0.016849
PI	0.329162
THOMPSON	1.383582
KG	0.016849
ENTROPY	2.314625
PORTFOLIO	0.975408

Iteration 15

Method	Max Value
UCB	1.346186
EI	0.016488
PI	0.327134
THOMPSON	1.420075
KG	0.016488
ENTROPY	2.280666
PORTFOLIO	0.955432

Iteration 16

Method	Max Value
UCB	1.345210
EI	0.016812
PI	0.341285
THOMPSON	1.409864
KG	0.016812
ENTROPY	2.224414
PORTFOLIO	0.964522

Iteration 17

Method	Max Value
UCB	1.344453
EI	0.016078

PI	0.332866
THOMPSON	1.433292
KG	0.016078
ENTROPY	2.254005
PORTFOLIO	0.952783

Iteration 18

Method	Max Value
UCB	1.343877
EI	0.015735
PI	0.331919
THOMPSON	1.358791
KG	0.015735
ENTROPY	2.195736
PORTFOLIO	0.939421

Iteration 19

Method	Max Value
UCB	1.343757
EI	0.015710
PI	0.340059
THOMPSON	1.393799
KG	0.015710
ENTROPY	2.142284
PORTFOLIO	0.950183

Iteration 20

Method	Max Value
UCB	1.343695
EI	0.015338
PI	0.331324
THOMPSON	1.389507
KG	0.015338
ENTROPY	2.153138
PORTFOLIO	0.934796

Iteration 21

Method	Max Value
UCB	1.334196
EI	0.014130
PI	0.327722
THOMPSON	1.369107
KG	0.014130
ENTROPY	2.105779
PORTFOLIO	0.926558

Iteration 22

Method	Max Value
UCB	1.329906
EI	0.012645
PI	0.312764
THOMPSON	1.346593
KG	0.012645
ENTROPY	2.133827
PORTFOLIO	0.921707

Iteration 23

Method	Max Value
UCB	1.329755
EI	0.012509
PI	0.318147
THOMPSON	1.391681
KG	0.012509
ENTROPY	2.102013
PORTFOLIO	0.918452

Iteration 24

Method	Max Value
UCB	1.332813
EI	0.013126

PI	0.325284
THOMPSON	1.369125
KG	0.013126
ENTROPY	2.083374
PORTFOLIO	0.923067

Iteration 25

Method	Max Value
UCB	1.322703
EI	0.008847
PI	0.289030
THOMPSON	1.389783
KG	0.008847
ENTROPY	2.084888
PORTFOLIO	0.912379

Iteration 26

Method	Max Value
UCB	1.326046
EI	0.009223
PI	0.279877
THOMPSON	1.351121
KG	0.009223
ENTROPY	2.079563
PORTFOLIO	0.908511

Iteration 27

Method	Max Value
UCB	1.326745
EI	0.008756
PI	0.291538
THOMPSON	1.328631
KG	0.008756
ENTROPY	2.111821
PORTFOLIO	0.899207

Iteration 28

Method	Max Value
UCB	1.319054
EI	0.007700
PI	0.264528
THOMPSON	1.344503
KG	0.007700
ENTROPY	2.043878
PORTFOLIO	0.899165

Iteration 29

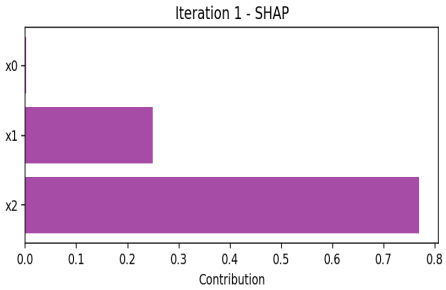
Method	Max Value
UCB	1.321345
EI	0.008051
PI	0.276152
THOMPSON	1.328945
KG	0.008051
ENTROPY	2.035299
PORTFOLIO	0.906438

Iteration 30

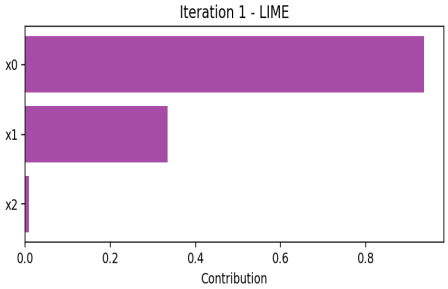
Method	Max Value
UCB	1.317725
EI	0.008061
PI	0.279255
THOMPSON	1.345441
KG	0.008061
ENTROPY	2.054032
PORTFOLIO	0.891219

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



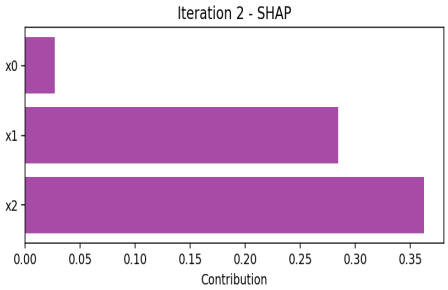
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.00114
x1	0.24874
x2	0.76897

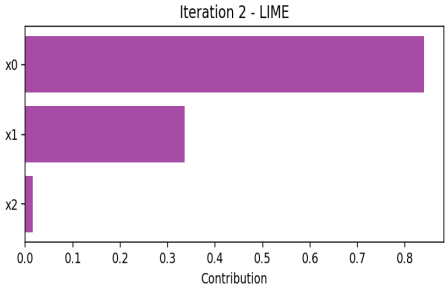
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.93648
x1	0.33375
x2	0.00828

Iteration 2 - SHAP



Iteration 2 - LIME



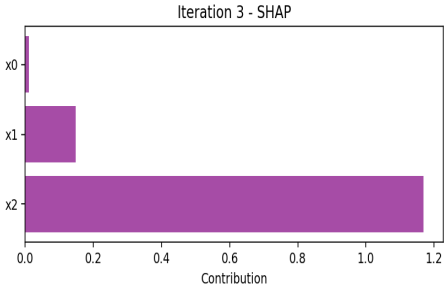
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.02661
x1	0.28428
x2	0.36214

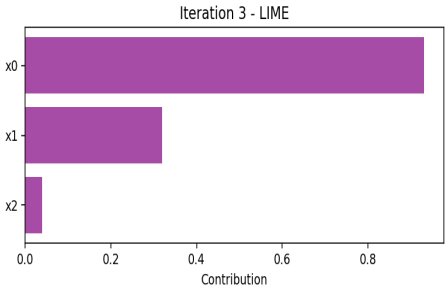
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.83992
x1	0.33551
x2	0.01606

Iteration 3 - SHAP



Iteration 3 - LIME



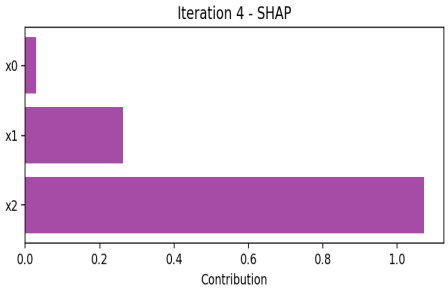
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.01036
x1	0.14767
x2	1.16933

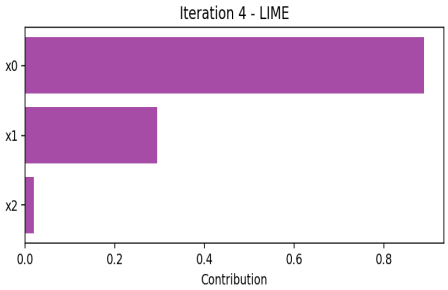
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.93100
x1	0.31971
x2	0.03954

Iteration 4 - SHAP



Iteration 4 - LIME



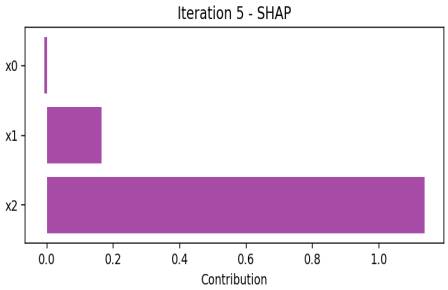
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.02890
x1	0.26329
x2	1.07160

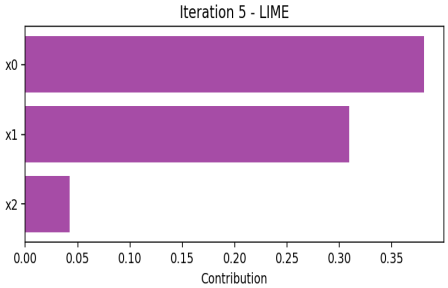
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.88902
x1	0.29360
x2	0.01913

Iteration 5 - SHAP



Iteration 5 - LIME



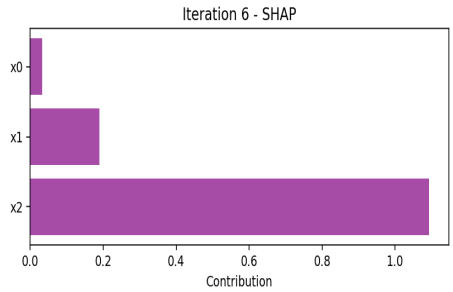
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.00742
x1	0.16551
x2	1.13824

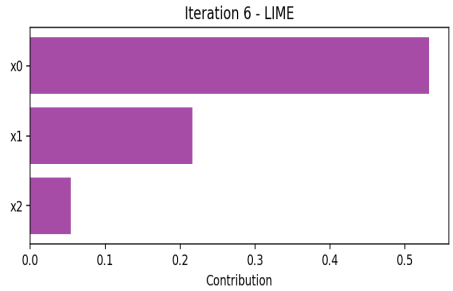
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.38080
x1	0.30955
x2	0.04216

Iteration 6 - SHAP



Iteration 6 - LIME



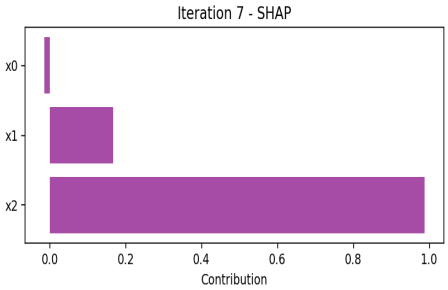
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	0.03171
x1	0.18932
x2	1.09258

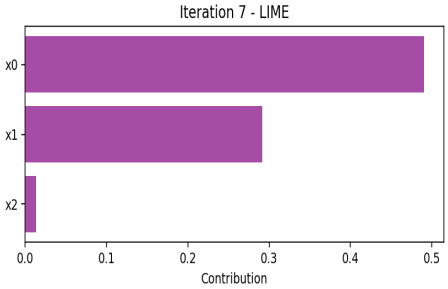
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.53181
x1	0.21653
x2	0.05407

Iteration 7 - SHAP



Iteration 7 - LIME



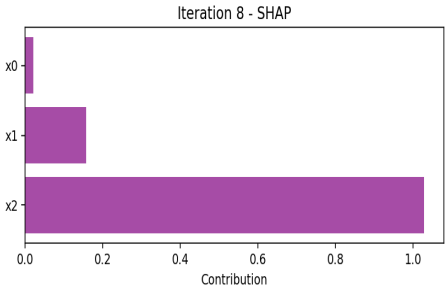
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.01406
x1	0.16758
x2	0.98815

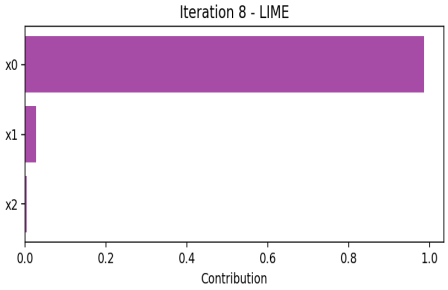
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.49033
x1	0.29144
x2	0.01294

Iteration 8 - SHAP



Iteration 8 - LIME



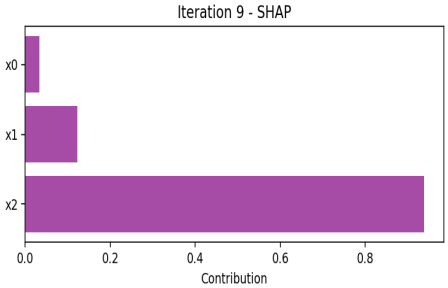
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.02131
x1	0.15756
x2	1.02805

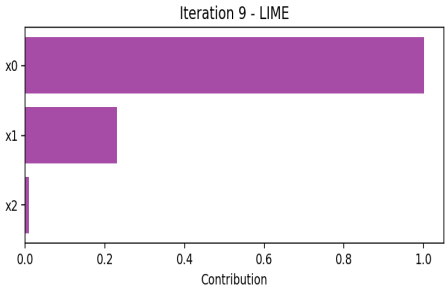
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.98542
x1	0.02655
x2	0.00332

Iteration 9 - SHAP



Iteration 9 - LIME



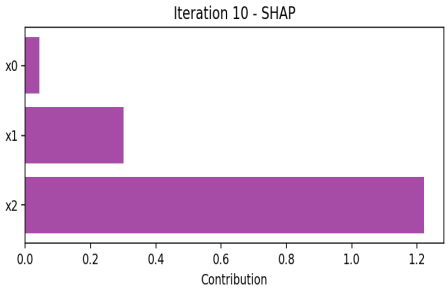
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.03301
x1	0.12211
x2	0.93765

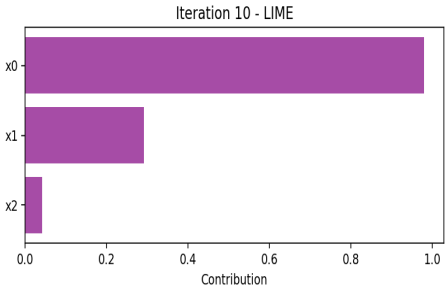
Iteration 9 - LIME Values

Feature	LIME Value
x0	1.00102
x1	0.22973
x2	0.00889

Iteration 10 - SHAP



Iteration 10 - LIME



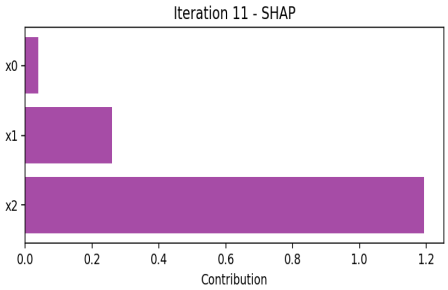
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.04241
x1	0.30035
x2	1.22059

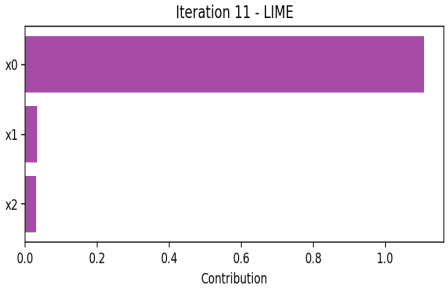
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.97910
x1	0.29115
x2	0.04147

Iteration 11 - SHAP



Iteration 11 - LIME



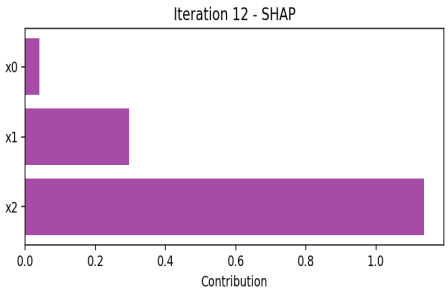
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	0.03917
x1	0.25934
x2	1.19276

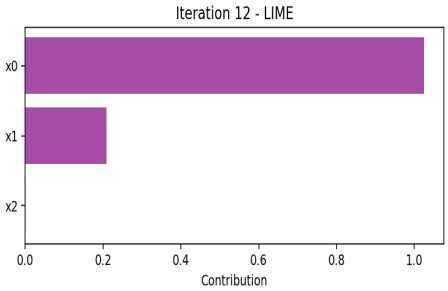
Iteration 11 - LIME Values

Feature	LIME Value
x0	1.10728
x1	0.03349
x2	0.03006

Iteration 12 - SHAP



Iteration 12 - LIME



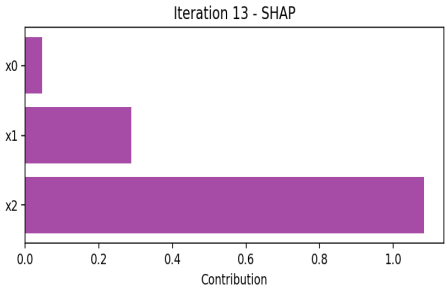
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	0.04078
x1	0.29656
x2	1.13639

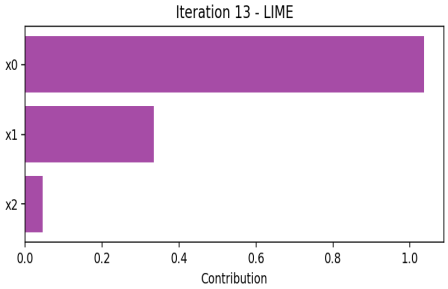
Iteration 12 - LIME Values

Feature	LIME Value
x0	1.02450
x1	0.20871
x2	0.00134

Iteration 13 - SHAP



Iteration 13 - LIME



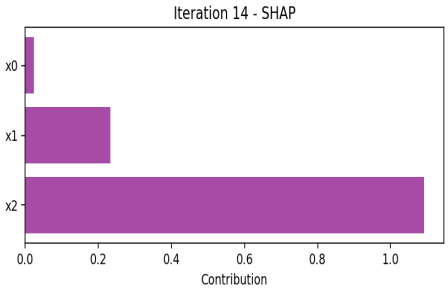
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.04555
x1	0.28802
x2	1.08352

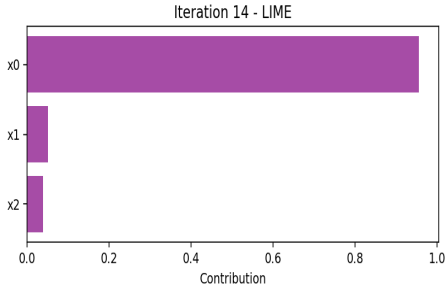
Iteration 13 - LIME Values

Feature	LIME Value
x0	1.03619
x1	0.33455
x2	0.04500

Iteration 14 - SHAP



Iteration 14 - LIME



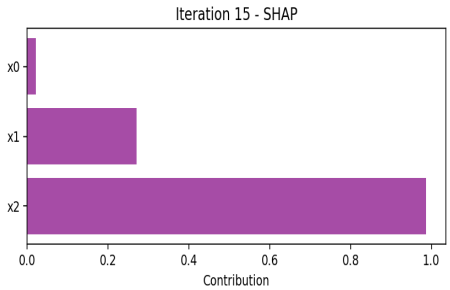
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.02369
x1	0.23306
x2	1.09075

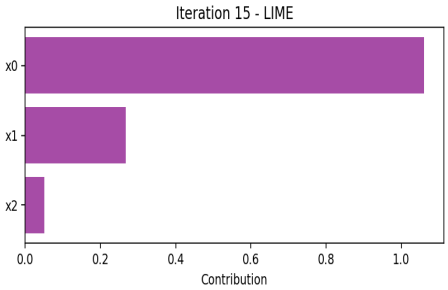
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.95577
x1	0.05016
x2	0.03862

Iteration 15 - SHAP



Iteration 15 - LIME



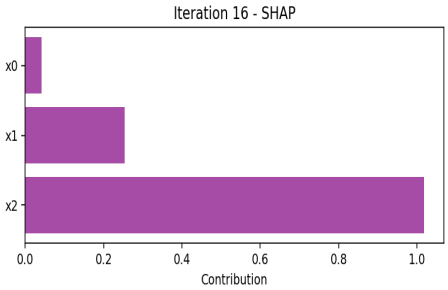
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.02143
x1	0.27047
x2	0.98658

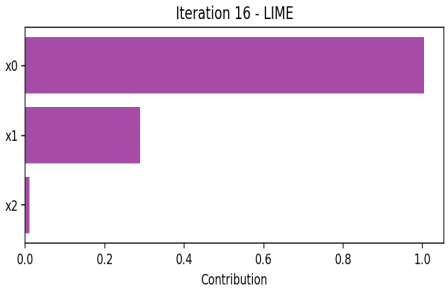
Iteration 15 - LIME Values

Feature	LIME Value
x0	1.05982
x1	0.26641
x2	0.05015

Iteration 16 - SHAP



Iteration 16 - LIME



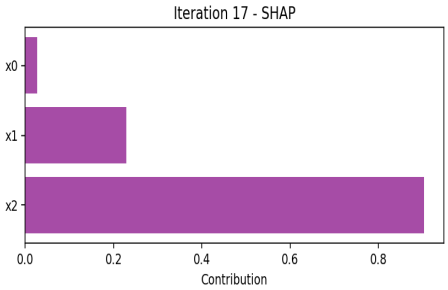
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	0.04108
x1	0.25340
x2	1.01779

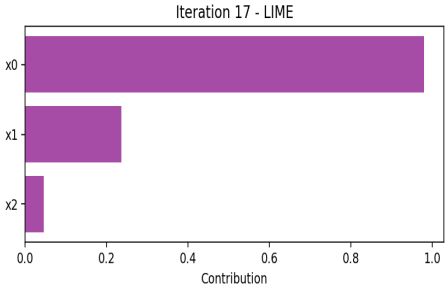
Iteration 16 - LIME Values

Feature	LIME Value
x0	1.00348
x1	0.28947
x2	0.01045

Iteration 17 - SHAP



Iteration 17 - LIME



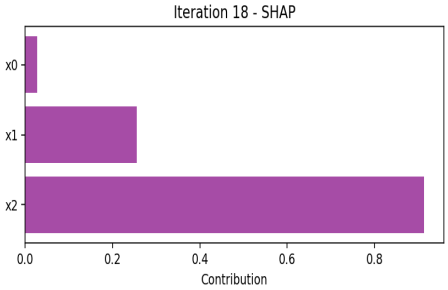
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	0.02656
x1	0.22933
x2	0.90327

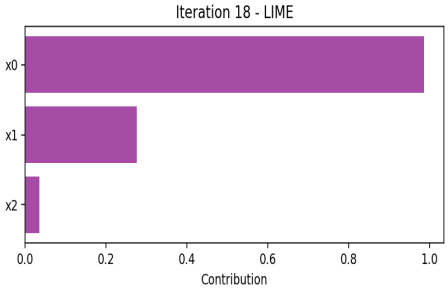
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.97937
x1	0.23550
x2	0.04585

Iteration 18 - SHAP



Iteration 18 - LIME



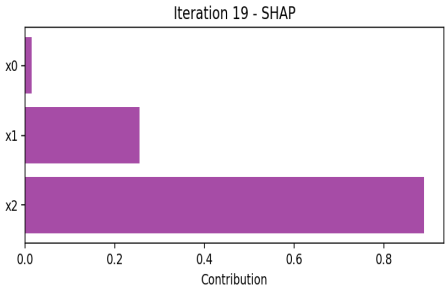
Iteration 18 - SHAP Values

Feature	SHAP Value
x0	0.02679
x1	0.25581
x2	0.91323

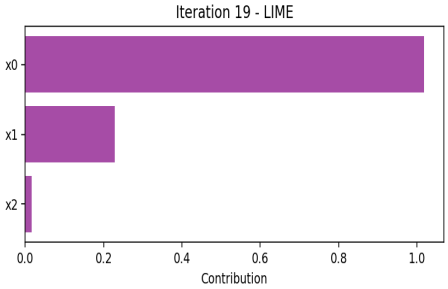
Iteration 18 - LIME Values

Feature	LIME Value
x0	0.98574
x1	0.27604
x2	0.03451

Iteration 19 - SHAP



Iteration 19 - LIME



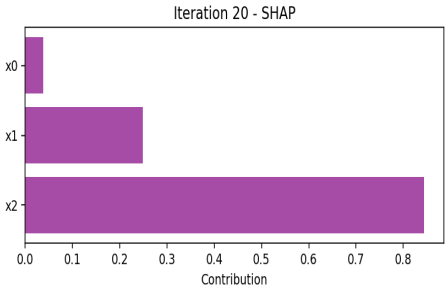
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	0.01405
x1	0.25432
x2	0.88952

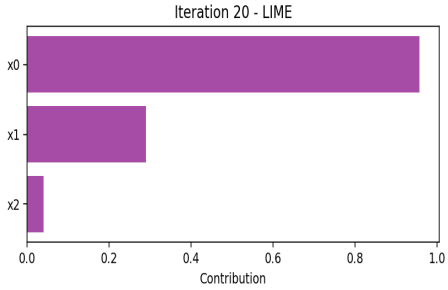
Iteration 19 - LIME Values

Feature	LIME Value
x0	1.01775
x1	0.22872
x2	0.01638

Iteration 20 - SHAP



Iteration 20 - LIME



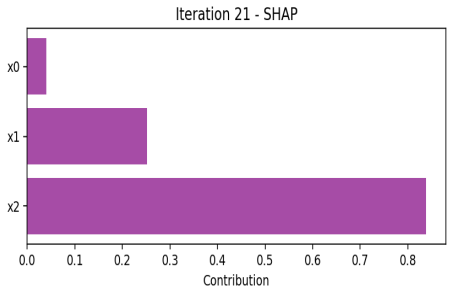
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	0.03780
x1	0.24848
x2	0.84369

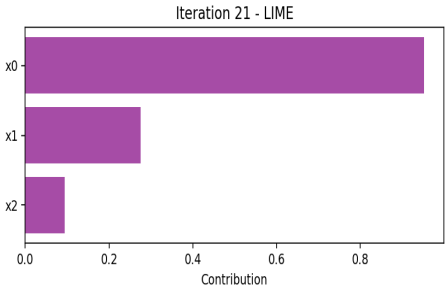
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.95737
x1	0.28904
x2	0.03980

Iteration 21 - SHAP



Iteration 21 - LIME



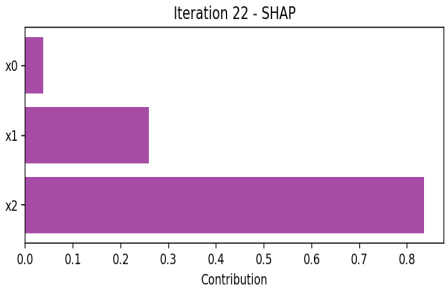
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	0.03938
x1	0.25154
x2	0.83790

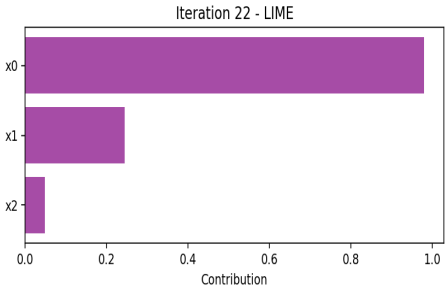
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.95138
x1	0.27513
x2	0.09366

Iteration 22 - SHAP



Iteration 22 - LIME



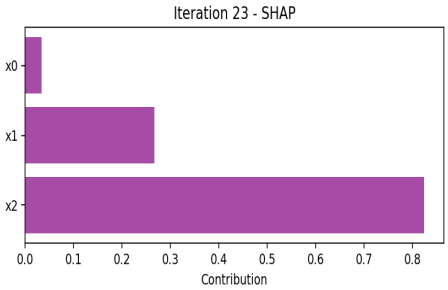
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	0.03710
x1	0.25842
x2	0.83507

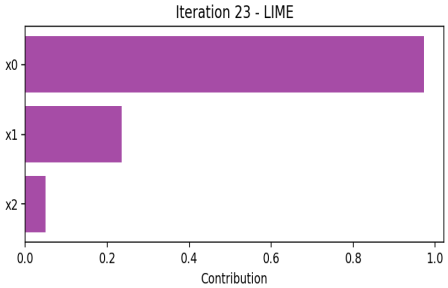
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.97866
x1	0.24458
x2	0.04802

Iteration 23 - SHAP



Iteration 23 - LIME



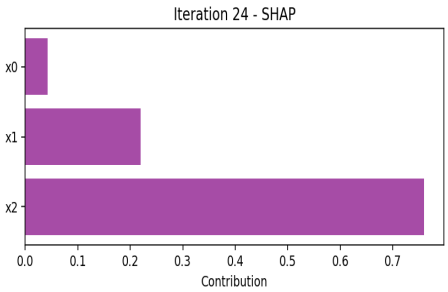
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	0.03347
x1	0.26670
x2	0.82369

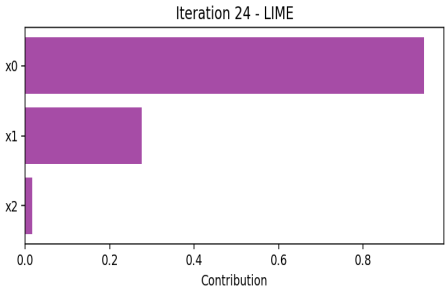
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.97219
x1	0.23454
x2	0.04966

Iteration 24 - SHAP



Iteration 24 - LIME



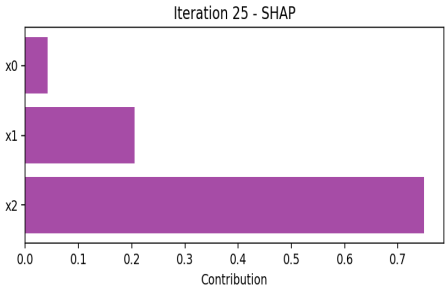
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	0.04247
x1	0.21917
x2	0.75920

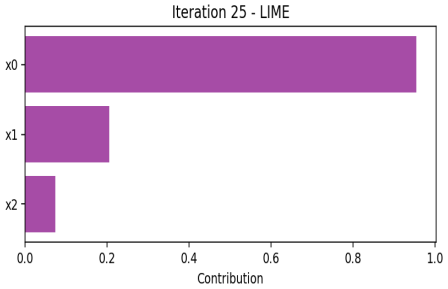
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.94368
x1	0.27565
x2	0.01590

Iteration 25 - SHAP



Iteration 25 - LIME



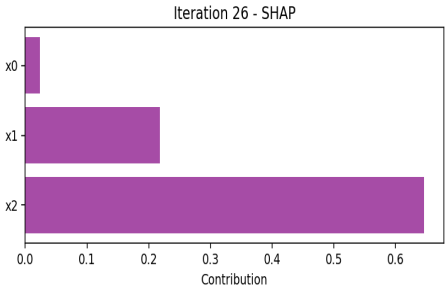
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	0.04222
x1	0.20491
x2	0.74885

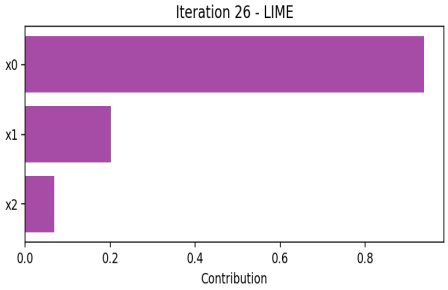
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.95411
x1	0.20510
x2	0.07308

Iteration 26 - SHAP



Iteration 26 - LIME



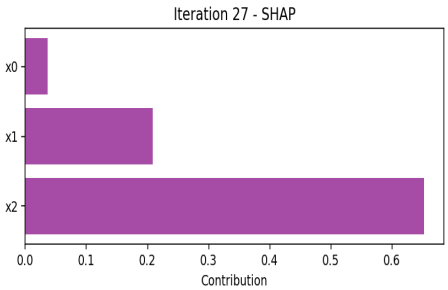
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	0.02407
x1	0.21808
x2	0.64602

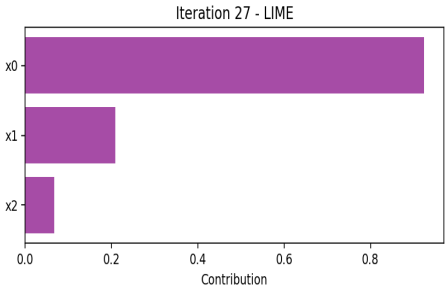
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.93821
x1	0.20162
x2	0.06840

Iteration 27 - SHAP



Iteration 27 - LIME



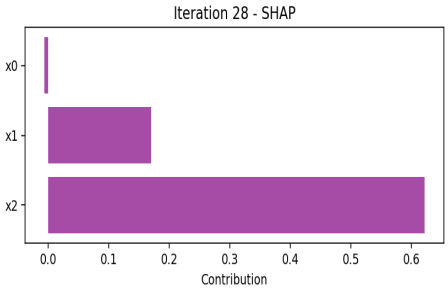
Iteration 27 - SHAP Values

Feature	SHAP Value
x0	0.03659
x1	0.20866
x2	0.65239

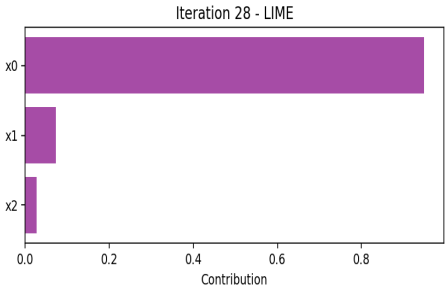
Iteration 27 - LIME Values

Feature	LIME Value
x0	0.92375
x1	0.20880
x2	0.06680

Iteration 28 - SHAP



Iteration 28 - LIME



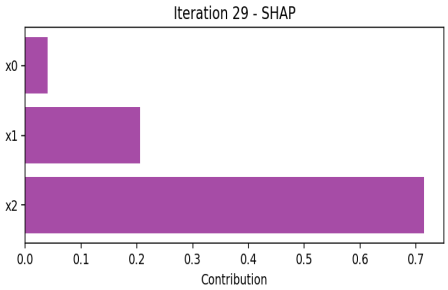
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	-0.00661
x1	0.16958
x2	0.62232

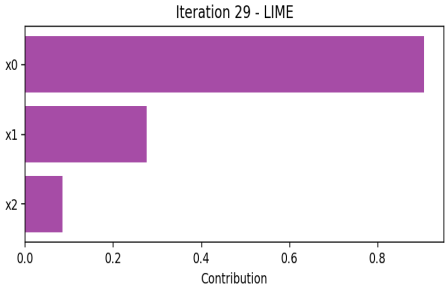
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.94858
x1	0.07278
x2	0.02650

Iteration 29 - SHAP



Iteration 29 - LIME



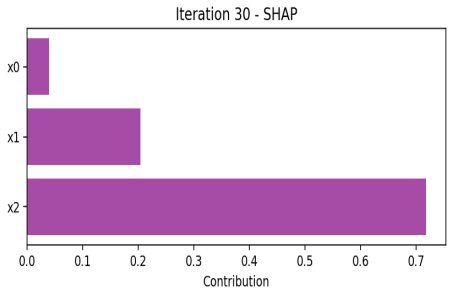
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	0.04019
x1	0.20610
x2	0.71443

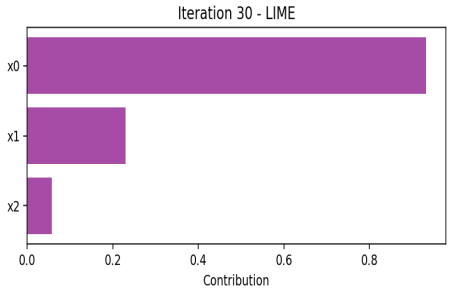
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.90520
x1	0.27543
x2	0.08407

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	0.03957
x1	0.20417
x2	0.71773

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.93169
x1	0.22992
x2	0.05766